- **Basic concept of Double integral, Double integral in Cartesian and Polar Coordinate system**

Prerequisite

READ:
HOWARD ANTON - 11.8: Cylindrical and Spherical Coordinates
Read only the selected parts from:
https://shorturl.at/sySX8

WATCH:
Pixar: The math behind the movies - Tony DeRose
Coordinate Systems: Animated
READ: Polar and Cartesian Coordinates or **WATCH:** **Intro to Polar Coordinates**
Sketching Polar Curves

CalcBLUE 3 : Ch. 13.5 : Cylindrical Coordinates
CalcBLUE 3 : Ch. 14.1 : Intro to Spherical Coordinates
CalcBLUE 3 : Ch. 14.2 : Visualizing Spherical Coordinates
Spherical Coordinate System (With 3D Animation)
Twos-Day Tuesday: 22 Slices of Pizza with Six Cuts

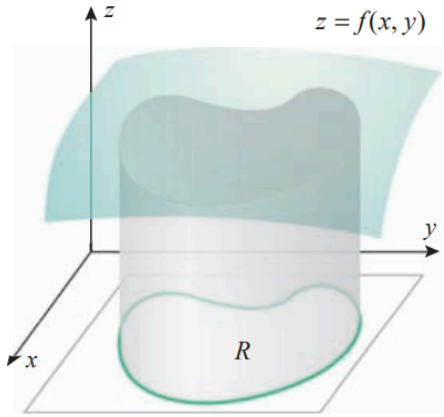
READ: HOWARD ANTON: 14.1: Double Integrals

VOLUME

Recall that the definite integral of a function of one variable

$$\int_a^b f(x)\,dx = \lim_{\max \Delta x_k \to 0} \sum_{k=1}^{n} f(x_k^*)\Delta x_k = \lim_{n \to +\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x_k \qquad (1)$$

arose from the problem of finding areas under curves. [In the rightmost expression in (1), we use the "limit as $n \to +\infty$" to encapsulate the process by which we increase the number of subintervals of $[a, b]$ in such a way that the lengths of the subintervals approach zero.] Integrals of functions of two variables arise from the problem of finding volumes under surfaces.



▲ **Figure 14.1.1**

14.1.1 THE VOLUME PROBLEM Given a function f of two variables that is continuous and nonnegative on a region R in the xy-plane, find the volume of the solid enclosed between the surface $z = f(x, y)$ and the region R (Figure 14.1.1).

Later, we will place more restrictions on the region R, but for now we will just assume that the entire region can be enclosed within some suitably large rectangle with sides parallel to the coordinate axes. This ensures that R does not extend indefinitely in any direction.

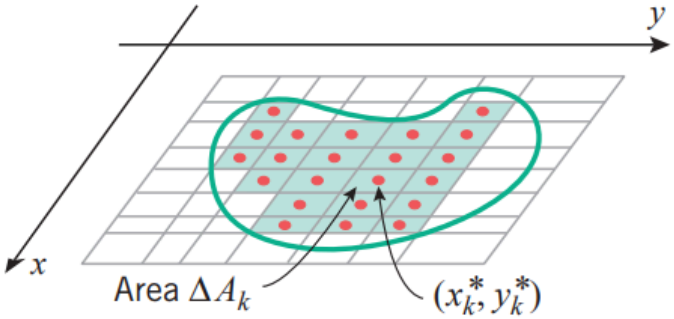
The procedure for finding the volume V of the solid in Figure 14.1.1 will be similar to the limiting process used for finding areas, except that now the approximating elements will be rectangular parallelepipeds rather than rectangles. We proceed as follows:

- Using lines parallel to the coordinate axes, divide the rectangle enclosing the region R into subrectangles, and exclude from consideration all those subrectangles that contain any points outside of R. This leaves only rectangles that are subsets of R (Figure 14.1.2). Assume that there are n such rectangles, and denote the area of the kth such rectangle by ΔA_k.
- Choose any arbitrary point in each subrectangle, and denote the point in the kth subrectangle by (x_k^*, y_k^*). As shown in Figure 14.1.3, the product $f(x_k^*, y_k^*)\Delta A_k$ is the volume of a rectangular parallelepiped with base area ΔA_k and height $f(x_k^*, y_k^*)$, so the sum

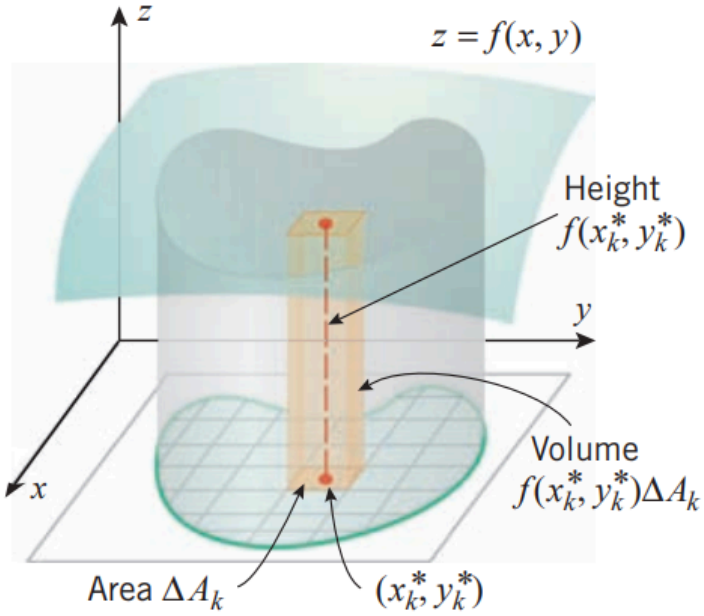
$$\sum_{k=1}^{n} f(x_k^*, y_k^*)\Delta A_k$$

  can be viewed as an approximation to the volume V of the entire solid.
- There are two sources of error in the approximation: first, the parallelepipeds have flat tops, whereas the surface $z = f(x, y)$ may be curved; second, the rectangles that form the bases of the parallelepipeds may not completely cover the region R. However, if we repeat the above process with more and more subdivisions in such a way that both the lengths and the widths of the subrectangles approach zero, then it is plausible that the errors of both types approach zero, and the exact volume of the solid will be

$$V = \lim_{n \to +\infty} \sum_{k=1}^{n} f(x_k^*, y_k^*)\Delta A_k$$



▲ **Figure 14.1.2**



▲ **Figure 14.1.3**

This suggests the following definition.

Definition 14.1.2 is satisfactory for our present purposes, but some issues would have to be resolved before it could be regarded as rigorous. For example, we would have to prove that the limit actually exists and that its value does not depend on how the points $(x_1^*, y_1^*), (x_2^*, y_2^*), \ldots, (x_n^*, y_n^*)$ are chosen. These facts are true if the region R is not too "complicated" and if f is continuous on R. The details are beyond the scope of this text.

14.1.2 DEFINITION (*Volume Under a Surface*) If f is a function of two variables that is continuous and nonnegative on a region R in the xy-plane, then the volume of the solid enclosed between the surface $z = f(x, y)$ and the region R is defined by

$$V = \lim_{n \to +\infty} \sum_{k=1}^{n} f(x_k^*, y_k^*) \Delta A_k \tag{2}$$

Here, $n \to +\infty$ indicates the process of increasing the number of subrectangles of the rectangle enclosing R in such a way that both the lengths and the widths of the subrectangles approach zero.

DEFINITION OF A DOUBLE INTEGRAL

As in Definition 14.1.2, the notation $n \to +\infty$ in (3) encapsulates a process in which the enclosing rectangle for R is repeatedly subdivided in such a way that both the lengths and the widths of the subrectangles approach zero. Note that subdividing so that the subrectangle lengths approach zero forces the mesh of the partition of the length of the enclosing rectangle for R to approach zero. Similarly, subdividing so that the subrectangle widths approach zero forces the mesh of the partition of the width of the enclosing rectangle for R to approach zero. Thus, we have extended the notion conveyed by Formula (1) where the definite integral of a one-variable function is expressed as a limit of Riemann sums. By extension, the sums in (3) are also called ***Riemann sums***, and the limit of the Riemann sums is denoted by

$$\iint_R f(x, y)\, dA = \lim_{n \to +\infty} \sum_{k=1}^{n} f(x_k^*, y_k^*) \Delta A_k \tag{4}$$

which is called the ***double integral*** of $f(x, y)$ over R.

If f is continuous and nonnegative on the region R, then the volume formula in (2) can be expressed as

$$V = \iint_R f(x, y)\, dA \tag{5}$$

If f has both positive and negative values on R, then a positive value for the double integral of f over R means that there is more volume above R than below, a negative value for the double integral means that there is more volume below R than above, and a value of zero means that the volume above R is the same as the volume below R.

EVALUATING DOUBLE INTEGRALS

Except in the simplest cases, it is impractical to obtain the value of a double integral from the limit in (4). However, we will now show how to evaluate double integrals by calculating

two successive single integrals. For the rest of this section we will limit our discussion to the case where R is a rectangle; in the next section we will consider double integrals over more complicated regions.

The partial derivatives of a function $f(x, y)$ are calculated by holding one of the variables fixed and differentiating with respect to the other variable. Let us consider the reverse of this process, ***partial integration***. The symbols

$$\int_a^b f(x, y)\,dx \quad \text{and} \quad \int_c^d f(x, y)\,dy$$

denote ***partial definite integrals***; the first integral, called the ***partial definite integral with respect to x***, is evaluated by holding y fixed and integrating with respect to x, and the second integral, called the ***partial definite integral with respect to y***, is evaluated by holding x fixed and integrating with respect to y. As the following example shows, the partial definite integral with respect to x is a function of y, and the partial definite integral with respect to y is a function of x.

▶ Example 1

$$\int_0^1 xy^2\,dx = y^2\int_0^1 x\,dx = \frac{y^2x^2}{2}\Big]_{x=0}^{1} = \frac{y^2}{2}$$

$$\int_0^1 xy^2\,dy = x\int_0^1 y^2\,dy = \frac{xy^3}{3}\Big]_{y=0}^{1} = \frac{x}{3}$$ ◀

A partial definite integral with respect to x is a function of y and hence can be integrated with respect to y; similarly, a partial definite integral with respect to y can be integrated with respect to x. This two-stage integration process is called ***iterated*** (or ***repeated***) ***integration***. We introduce the following notation:

$$\int_c^d \int_a^b f(x, y)\,dx\,dy = \int_c^d \left[\int_a^b f(x, y)\,dx\right] dy \tag{6}$$

$$\int_a^b \int_c^d f(x, y)\,dy\,dx = \int_a^b \left[\int_c^d f(x, y)\,dy\right] dx \tag{7}$$

These integrals are called ***iterated integrals***.

► **Example 2** Evaluate

$$\text{(a)} \int_1^3 \int_2^4 (40 - 2xy)\, dy\, dx \qquad \text{(b)} \int_2^4 \int_1^3 (40 - 2xy)\, dx\, dy$$

Solution (a).

$$\begin{aligned}
\int_1^3 \int_2^4 (40 - 2xy)\, dy\, dx &= \int_1^3 \left[\int_2^4 (40 - 2xy)\, dy \right] dx \\
&= \int_1^3 (40y - xy^2)\Big]_{y=2}^4 dx \\
&= \int_1^3 [(160 - 16x) - (80 - 4x)]\, dx \\
&= \int_1^3 (80 - 12x)\, dx \\
&= (80x - 6x^2)\Big]_1^3 = 112
\end{aligned}$$

Solution (b).

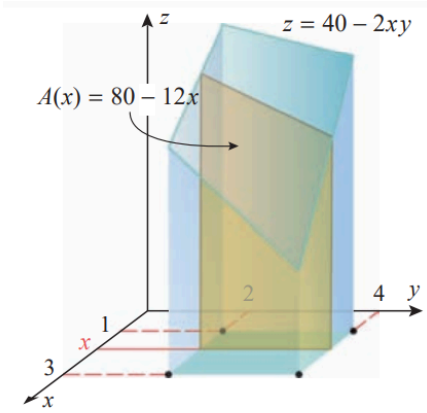
$$\int_2^4 \int_1^3 (40 - 2xy)\,dx\,dy = \int_2^4 \left[\int_1^3 (40 - 2xy)\,dx\right] dy$$

$$= \int_2^4 (40x - x^2y)\Big]_{x=1}^{3}\,dy$$

$$= \int_2^4 [(120 - 9y) - (40 - y)]\,dy$$

$$= \int_2^4 (80 - 8y)\,dy$$

$$= (80y - 4y^2)\Big]_2^4 = 112 \quad \blacktriangleleft$$



▲ **Figure 14.1.4**

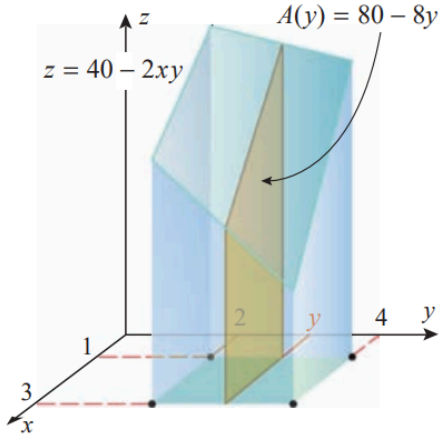
It is no accident that both parts of Example 2 produced the same answer. Consider the solid S bounded above by the surface $z = 40 - 2xy$ and below by the rectangle R defined by $1 \le x \le 3$ and $2 \le y \le 4$. By the method of slicing discussed in Section 6.2, the volume of S is given by

$$V = \int_1^3 A(x)\,dx$$

where $A(x)$ is the area of a vertical cross section of S taken perpendicular to the x-axis (Figure 14.1.4). For a fixed value of x, $1 \le x \le 3$, $z = 40 - 2xy$ is a function of y, so the integral

$$A(x) = \int_2^4 (40 - 2xy)\,dy$$

represents the area under the graph of this function of y. Thus,



▲ **Figure 14.1.5**

represents the area under the graph of this function of y. Thus,

$$V = \int_1^3 \left[\int_2^4 (40 - 2xy)\, dy \right] dx = \int_1^3 \int_2^4 (40 - 2xy)\, dy\, dx$$

is the volume of S. Similarly, by the method of slicing with cross sections of S taken perpendicular to the y-axis, the volume of S is given by

$$V = \int_2^4 A(y)\, dy = \int_2^4 \left[\int_1^3 (40 - 2xy)\, dx \right] dy = \int_2^4 \int_1^3 (40 - 2xy)\, dx\, dy$$

(Figure 14.1.5). Thus, the iterated integrals in parts (a) and (b) of Example 2 both measure the volume of S, which by Formula (5) is the double integral of $z = 40 - 2xy$ over R. That is,

$$\int_1^3 \int_2^4 (40 - 2xy)\, dy\, dx = \iint_R (40 - 2xy)\, dA = \int_2^4 \int_1^3 (40 - 2xy)\, dx\, dy$$

We will often denote the rectangle

$$\{(x, y) : a \le x \le b, c \le y \le d\}$$

as $[a, b] \times [c, d]$ for simplicity.

The geometric argument above applies to any continuous function $f(x, y)$ that is nonnegative on a rectangle $R = [a, b] \times [c, d]$, as is the case for $f(x, y) = 40 - 2xy$ on $[1, 3] \times [2, 4]$. The conclusion that the double integral of $f(x, y)$ over R has the same value as either of the two possible iterated integrals is true even when f is negative at some points in R. We state this result in the following theorem and omit a formal proof.

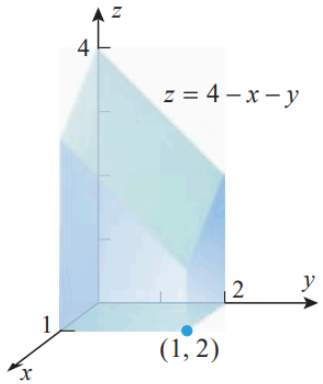
14.1.3 THEOREM (*Fubini's Theorem*) *Let R be the rectangle defined by the inequalities*

$$a \le x \le b, \quad c \le y \le d$$

If $f(x, y)$ is continuous on this rectangle, then

$$\iint_R f(x, y)\, dA = \int_c^d \int_a^b f(x, y)\, dx\, dy = \int_a^b \int_c^d f(x, y)\, dy\, dx$$

Theorem 14.1.3 allows us to evaluate a double integral over a rectangle by converting it to an iterated integral. This can be done in two ways, both of which produce the value of the double integral.



▲ **Figure 14.1.6**

TECHNOLOGY MASTERY

If you have a CAS with a built-in capability for computing iterated double integrals, use it to check Example 3.

► **Example 3** Use a double integral to find the volume of the solid that is bounded above by the plane $z = 4 - x - y$ and below by the rectangle $R = [0, 1] \times [0, 2]$ (Figure 14.1.6).

Solution. The volume is the double integral of $z = 4 - x - y$ over R. Using Theorem 14.1.3, this can be obtained from either of the iterated integrals

$$\int_0^2 \int_0^1 (4 - x - y)\, dx\, dy \quad \text{or} \quad \int_0^1 \int_0^2 (4 - x - y)\, dy\, dx \tag{8}$$

Using the first of these, we obtain

$$V = \iint_R (4 - x - y)\, dA = \int_0^2 \int_0^1 (4 - x - y)\, dx\, dy$$

$$= \int_0^2 \left[4x - \frac{x^2}{2} - xy\right]_{x=0}^{1} dy = \int_0^2 \left(\frac{7}{2} - y\right) dy$$

$$= \left[\frac{7}{2}y - \frac{y^2}{2}\right]_0^2 = 5$$

You can check this result by evaluating the second integral in (8). ◄

Theorem 14.1.3 guarantees that the double integral in Example 4 can also be evaluated by integrating first with respect to y and then with respect to x. Verify this.

► **Example 4** Evaluate the double integral

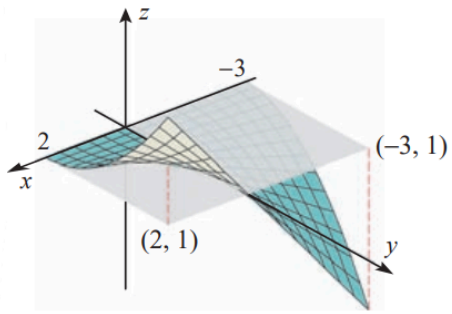
$$\iint_R y^2 x \, dA$$

over the rectangle $R = \{(x, y) : -3 \le x \le 2, 0 \le y \le 1\}$.

Solution. In view of Theorem 14.1.3, the value of the double integral can be obtained by evaluating one of two possible iterated double integrals. We choose to integrate first with respect to x and then with respect to y.

$$\iint_R y^2 x \, dA = \int_0^1 \int_{-3}^2 y^2 x \, dx \, dy = \int_0^1 \left[\frac{1}{2} y^2 x^2\right]_{x=-3}^2 dy$$

$$= \int_0^1 \left(-\frac{5}{2} y^2\right) dy = -\frac{5}{6} y^3 \Big]_0^1 = -\frac{5}{6} \quad ◄$$



$z = y^2 x$ on $[-3, 2] \times [0, 1]$

▲ **Figure 14.1.7**

The integral in Example 4 can be interpreted as the net signed volume between the rectangle $[-3, 2] \times [0, 1]$ and the surface $z = y^2 x$. That is, it is the volume below $z = y^2 x$ and above $[0, 2] \times [0, 1]$ minus the volume above $z = y^2 x$ and below $[-3, 0] \times [0, 1]$ (Figure 14.1.7).

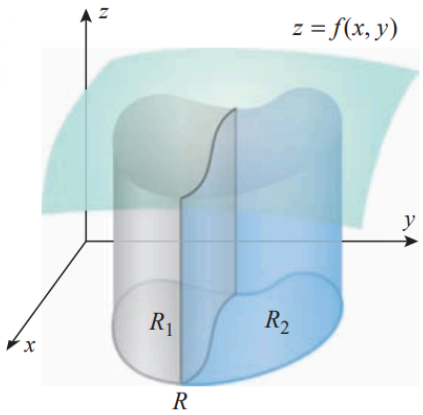
PROPERTIES OF DOUBLE INTEGRALS

To distinguish between double integrals of functions of two variables and definite integrals of functions of one variable, we will refer to the latter as ***single integrals***. Because double integrals, like single integrals, are defined as limits, they inherit many of the properties of limits. The following results, which we state without proof, are analogs of those in Theorem 5.5.4.

$$\iint_R cf(x, y)\,dA = c\iint_R f(x, y)\,dA \quad (c \text{ a constant}) \tag{9}$$

$$\iint_R [f(x, y) + g(x, y)]\,dA = \iint_R f(x, y)\,dA + \iint_R g(x, y)\,dA \tag{10}$$

$$\iint_R [f(x, y) - g(x, y)]\,dA = \iint_R f(x, y)\,dA - \iint_R g(x, y)\,dA \tag{11}$$



The volume of the entire solid is the sum of the volumes of the solids above R_1 and R_2.

▲ **Figure 14.1.8**

It is evident intuitively that if $f(x, y)$ is nonnegative on a region R, then subdividing R into two regions R_1 and R_2 has the effect of subdividing the solid between R and $z = f(x, y)$ into two solids, the sum of whose volumes is the volume of the entire solid (Figure 14.1.8). This suggests the following result, which holds even if f has negative values:

$$\iint_R f(x, y)\,dA = \iint_{R_1} f(x, y)\,dA + \iint_{R_2} f(x, y)\,dA \tag{12}$$

The proof of this result will be omitted.

WATCH:

- **CalcBLUE 3 : Ch. 1.1 : Integrals, Definite & Indefinite**
- **CalcBLUE 3 : Ch. 1.2 : Recalling Riemann Sums**
- **CalcBLUE 3 : Ch. 1.3 : The Definite Integral**
- **CalcBLUE 3 : Ch. 1.4 : Multiple Integrals - What & Why?**
- **CalcBLUE 3 : Ch. 1.5 : Integrating Constants**
- **CalcBLUE 3 : Ch. 1.6 : BONUS! Integration Theory**

- **CalcBLUE 3 : Ch. 2.1 : What Do We Do Now?**
- **CalcBLUE 3 : Ch. 2.2 : The Fubini Theorem**
- **CalcBLUE 3 : Ch. 2.3 : Iterated Integrals**
- **CalcBLUE 3 : Ch. 2.4 : Fubini & Areas**

- **CalcBLUE 3 : Ch. 3.1 : Limits & Double Integrals**
- **CalcBLUE 3 : Ch. 3.2 : Examples - Simple Double Integrals**
- **CalcBLUE 3 : Ch. 3.3 : Example - Order of Integration**
- **CalcBLUE 3 : Ch. 3.4 : Example - Additivity of Integrals**
- **CalcBLUE 3 : Ch. 3.5 : Example - Improper, Unbounded Integrals**
- **Double integrals and Polar integrals: Explained with 3D visualizations**
- **How to Set Up Double Integrals**
- **Defining Double Integration with Riemann Sums | Volume under a Surface**
- **What are double integrals? What are they for? | #SoME1 #3b1b**

Use the following online solver:

Double Integral Calculator

Double Integral Calculator

Multiple (Double, Triple) Integral Calculator - eMathHelp

PRACTISE: HOWARD ANTON: 14.1: Double Integrals: 1-16

1–12 Evaluate the iterated integrals. ■

1. $\int_0^1 \int_0^2 (x+3)\,dy\,dx$

2. $\int_1^3 \int_{-1}^1 (2x-4y)\,dy\,dx$

3. $\int_2^4 \int_0^1 x^2 y\,dx\,dy$

4. $\int_{-2}^0 \int_{-1}^2 (x^2+y^2)\,dx\,dy$

5. $\int_0^{\ln 3} \int_0^{\ln 2} e^{x+y}\,dy\,dx$

6. $\int_0^2 \int_0^1 y \sin x\,dy\,dx$

7. $\int_{-1}^0 \int_2^5 dx\,dy$

8. $\int_4^6 \int_{-3}^7 dy\,dx$

9. $\int_0^1 \int_0^1 \frac{x}{(xy+1)^2}\,dy\,dx$

10. $\int_{\pi/2}^{\pi} \int_1^2 x \cos xy\,dy\,dx$

11. $\int_0^{\ln 2} \int_0^1 xye^{y^2x}\,dy\,dx$

12. $\int_3^4 \int_1^2 \frac{1}{(x+y)^2}\,dy\,dx$

13–16 Evaluate the double integral over the rectangular region R. ■

13. $\iint_R 4xy^3\,dA;\ R = \{(x, y) : -1 \le x \le 1, -2 \le y \le 2\}$

14. $\iint_R \dfrac{xy}{\sqrt{x^2 + y^2 + 1}}\,dA;$

$R = \{(x, y) : 0 \le x \le 1, 0 \le y \le 1\}$

15. $\iint_R x\sqrt{1 - x^2}\,dA;\ R = \{(x, y) : 0 \le x \le 1, 2 \le y \le 3\}$

16. $\iint_R (x \sin y - y \sin x)\,dA;$

$R = \{(x, y) : 0 \le x \le \pi/2, 0 \le y \le \pi/3\}$